

Warehouse Slotting Efficiency Analysis

Mapping Fast-Moving Parts Against the Shipping Dock

Prashanth Kathiresan

[linkedin.com/in/prashanth-kathiresan](https://www.linkedin.com/in/prashanth-kathiresan)

Stack: Epicor ERP (BAQ) · Power BI (Power Query / M, DAX) · Python visual (matplotlib)

EXECUTIVE SUMMARY



Warehouse-floor heatmap — each cell is a physical bin, colored by the count of distinct Smooth-demand parts. Shipping (green) is at the G bin, lower-left.

THE QUESTION Are our fast-moving (Smooth-demand) parts stored close to the shipping dock, or are they slotted in ways that make pickers walk farther than they should?

THE FINDING No. The heat sits **farthest from shipping** — densest in the right-side column and upper-left rack, while the bins beside the dock sit empty. Because Smooth parts are picked at a near-uniform rate, more parts in a bin means more trips to it. The most-picked bins are the longest walk away.

THE RECOMMENDATION Re-slot the hottest far bins into the open near-dock bins, **highest-leverage first**. Pilot the top two, re-pull the data after 30 days, and use the same heatmap as before/after proof.

THE STACK Epicor ERP (BAQ) → Power Query / M (clean & normalize bins) → data model + DAX (DISTINCTCOUNT) → custom Python heatmap inside Power BI.

Full methodology and analysis follow on the next page.

1 Summary

A manufacturing warehouse ships from a single dock located at the **Shipping** in the lower-left of the floor. In an efficient layout, the parts picked most often for shipment should sit closest to that dock. I built a warehouse-floor heatmap to test whether that was true for the facility's **Smooth-demand** parts — the segment of part numbers with steady, predictable pick activity.

It was not true. The heatmap showed that the variety of Smooth parts is concentrated farthest from shipping — densest in the right-side storage column and the upper-left rack — while the bins immediately around the dock sit nearly empty. Because Smooth parts share a narrow, near-uniform pick rate by definition, a bin holding more of them is a bin that gets visited more often. The current layout therefore forces pickers to travel the length of the building on the orders they fill most frequently.

I recommended re-slotting the densest far bins into the open bins beside shipping, starting with the two highest-leverage locations, and re-measuring with the same heatmap after 30 days to confirm the heat shifts toward the dock.

2 Ask — Defining the question

Business task: Are the warehouse's fast-moving (Smooth-demand) part numbers slotted efficiently relative to the shipping dock, or are they concentrated in locations that lengthen pick-to-ship travel?

Why it matters: Picking labor is one of the largest recurring costs in a warehouse, and travel distance is the single biggest driver of pick time. Fast-moving parts are, by definition, the ones touched most often, so where they live has an outsized effect on daily labor. If those parts sit far from the dock, every shift pays a travel penalty that compounds across thousands of picks.

Primary stakeholder: The warehouse / operations manager, who owns slotting decisions and can act on the result directly.

Success criterion: A clear, at-a-glance answer to “where do our fast movers live relative to shipping?” plus a ranked, actionable list of which bins to relocate first.

3 Prepare — Sourcing and qualifying the data

Source system. All data originates in the company's Epicor ERP, pulled through a Business Activity Query (BAQ) — Epicor's native query layer. The BAQ joins the part-bin placement records to the part master so that every row represents one part number in one physical bin, carrying its on-hand quantity, warehouse code, and bin location.

The data model. The work uses a two-table star schema:

- **Fact table** of part-to-bin placements (one row per part, per bin), holding the bin code, a normalized bin label, on-hand quantity, warehouse, and the part number.
- **Part-classification dimension** holding each part's demand segment and ABC value classifications.

The two tables relate one-to-many on **Part** — the classification dimension on the “one” side, the placement fact on the “many” side. This lets the analysis filter physical placements by demand characteristics that live in a separate table, which is closer to real analytical modeling than working from a single flat extract.

Defining “fast-moving.” Parts were classified into demand segments using a pick-variability method applied to historical pick activity. The Smooth segment captures parts with steady, low-variance demand — pick rates clustered tightly around the category average rather than spiking or going dormant. This project analyzes only the Smooth segment. The exact classification rule is proprietary to the company; it is described here at the level of method, not formula.

Credibility (ROCCC). The data is Reliable and Original — first-party, transaction-level ERP data, not a third-party estimate. It is Current, reflecting live bin placements at extract time, and Cited to a single authoritative source (Epicor). It is reasonably Comprehensive for the question, covering every Smooth part in the main warehouse. The main credibility caveat is scope, addressed under Limitations.

Data sensitivity and anonymization. Because the data comes from a live ERP, the real part numbers, descriptions, and counts are proprietary and cannot be published. For this public portfolio version, the identifying fields are replaced with a randomized representative sample that preserves the shape of the real distribution — the same concentration pattern and totals — without exposing any real part, customer, or volume data. The methodology shown is real; the figures shown are a sanitized stand-in. This is what makes the project shareable without disclosing company data.

Data dictionary (analysis fields).

Field	Table	Meaning
Part	Fact + Dimension	Part number; the join key and the unit being counted
Bin-Location (normalized)	Fact	Cleaned bin label (sub-bins collapsed to a parent bin) used to position each cell on the floor map
Demand Segment	Dimension	Demand classification; this analysis filters to “Smooth”
On Hand	Fact	Quantity on hand (not used here; we count parts, not quantity)
Warehouse	Fact	Warehouse code; filtered to the main facility

4 Process — Cleaning, modeling, and verifying

Bin normalization (Power Query / M). Raw ERP bin codes are messy: a single physical bin appears as many sub-locations (for example K-01, K-02, AS-3). Left as-is, the heatmap would fragment one shelf into a dozen cells. I normalized every bin code to its parent location — collapsing anything after the separator, handling the AS-prefixed bins and leading-zero codes as special cases — so that each physical bin maps to exactly one cell on the floor map. This logic was authored at the query layer (BAQ) and mirrored in Power Query, so the cleaning step is transparent and reproducible at both layers.

Filtering. Non-storage and transit locations (docks, holds, in-transit, inspection) were excluded, and the data was restricted to the main warehouse and to the Smooth demand segment.

The relationship. The placement fact connects to the classification dimension one-to-many on Part. Filtering Demand Segment = “Smooth” on the dimension side flows through the relationship to return only Smooth placements in the fact, which is what the visual counts.

Key measure (DAX). The heat value is a distinct count of part numbers per bin — how many unique Smooth parts live in each location. Counting distinct parts (rather than rows or quantity) is deliberate: the question is about how many kinds of fast movers a picker must visit a bin for, not how many units sit on the shelf.

Verifying integrity. The published total reconciles to a known control number, and the per-bin counts sum to that total — a simple but essential check that catches double-counting from the bin normalization or the join. The distinct-count logic was confirmed against a small manual sample before trusting the full visual.

5 Analyze — What the data shows

The core relationship: heat concentrates away from shipping. With shipping fixed at the G bin (lower-left), the heatmap reveals that Smooth parts are densest in the locations farthest from the dock. The hottest cells on the entire floor sit in the right-side storage column, and a secondary hot zone runs through the upper-left rack. The bins physically closest to shipping — the A–G letter row and the lowest rack positions — are empty or hold a single part.

Why distinct-part-count is a valid proxy for pick activity. This is the analytical hinge of the project. The Smooth segment is defined as parts with steady, low-variance demand clustered near the category average — so within this set, pick frequency is genuinely near-uniform, not merely assumed. When velocity is held roughly constant across every part, the number of Smooth parts in a bin becomes a reasonable proxy for how often that bin is picked. A bin with more Smooth parts is a bin a picker visits more often. This allows a part-count heatmap to stand in for a pick-frequency analysis without per-line transaction data — the uniformity is built into the segment definition rather than imposed on it.

The implication. Because the densest bins are also the farthest from shipping, the layout systematically routes pickers to the back of the building on the orders they fill most frequently. The cost is not in any single pick — it is in repeating a long pick path across every shift, every day.

Stated assumption. The proxy holds only as long as the Smooth classification is current. If parts that are no longer steady-movers remain tagged Smooth, the proxy weakens. The recommendation accounts for this by including periodic re-validation of the segment.

6 Share — The story on the dashboard

The dashboard is built to deliver one finding without explanation. The centerpiece is a custom warehouse-floor heatmap, rendered with a Python (matplotlib) visual inside Power BI because Power BI's native visuals cannot reproduce a true-to-life floor plan — bins are positioned to mirror their real physical location, not arranged on a generic grid. A green SHIPPING marker anchors the dock at the G bin so the spatial story is unmistakable. A companion table lists each bin and its part count for readers who want the exact numbers.

The narrative the report tells, top to bottom:

- 1. Context:** Shipping operates near to the G bin. Efficient layouts keep fast movers close to the dock.
- 2. Finding:** The heat — the concentration of Smooth parts — sits farthest from Shipping, in the right-side column and the upper-left rack. The bins around shipping are empty.

3. **So what:** Since Smooth parts are picked at a steady, comparable rate, the most-picked variety of parts is the farthest to walk to. The layout fights efficiency.

Why a Python visual here, and native elsewhere. The floor map is the one element that genuinely requires Python — a spatial layout no native chart supports. The supporting table and color legend are native Power BI, which handle those jobs well. Choosing Python only where it adds something native cannot is a deliberate design decision, not a default.

7 Act — Recommendations

Primary recommendation: re-slot Smooth parts from the far bins into the open bins beside shipping.

1. **Start with the highest-leverage bins.** The hottest cells in the right-side column are both the densest and among the farthest from G, so relocating them yields the largest travel reduction per part moved. These are the first targets.
2. **Use the open near-dock bins as the destination.** The bins immediately around shipping (the A–G letter row and the lowest rack positions) are already empty. The valuable real estate nearest the dock is sitting unused, so the relocation needs no new space.
3. **Pilot, then measure.** Re-slot the top two far-and-hot bins into the open near-dock bins, then re-pull the BAQ after 30 days and regenerate the same heatmap. If the heat has shifted toward shipping, the change worked — and the before/after heatmap is the proof.
4. **Keep the classification current.** Because the analysis rests on the Smooth segment being accurate, re-run the demand classification periodically so slotting decisions track real, current demand.

Expected impact. Because each Smooth part is picked at a similar steady rate, travel saved scales directly with the number of fast-moving parts moved closer to the dock. The hottest, farthest bins are therefore the highest-return relocation targets — the analysis doesn't just say "fix the layout," it ranks exactly where to start.

8 What this project demonstrates

- **End-to-end ownership:** sourcing from a live ERP via BAQ, modeling a two-table relationship, cleaning messy real-world identifiers, and delivering a decision-ready visual.
- **Tool judgment:** native Power BI where it fits, a custom Python visual only where native cannot reproduce a physical floor plan.
- **Analytical honesty:** an assumption (uniform velocity) grounded in the data's own definition rather than asserted, with its limits stated plainly.
- **Action, not just analysis:** a ranked recommendation and a built-in measurement loop, not a one-time chart.